

Likelihood-free phylogenetics



Luc Blassel

Auditions CNRS 51/02

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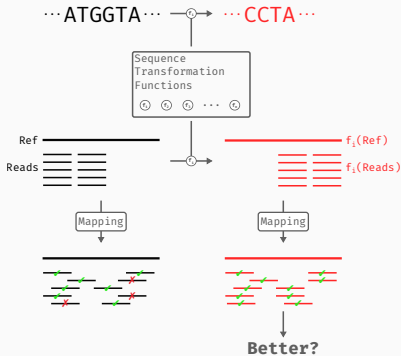
slides: lucblassel.com/CNRS_2026.pdf

Professional history

1. Ingénieur @ **AgroParisTech** 2014-2018
 - Specialisation in **ML** and **data-science**
 - Double diploma with the **IASD** MSc. @ **Dauphine**
2. PhD. @ **Institut Pasteur** 2018-2022
(5 papers)
 - *R. Chikhi*: **Improving** sequence **alignements**
 - *O. Gascuel*: **Learning** from **alignments**
3. Post-doc @ **LBBE & CQSB** 2023-
(1 paper & 2 preprints)
 - *L. Jacob & B. Boussau*:
Likelihood-free phylogenetic **inference**

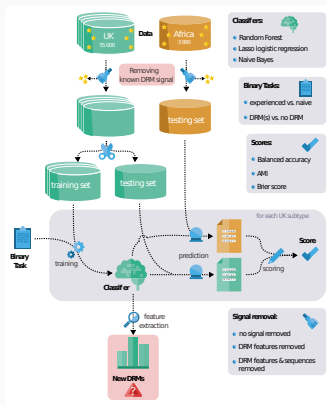
Previous works - PhD

keywords: Sequence Bioinformatics, combinatorics



- **Goal:** Improve mapping of NanoPore reads to ref. genome
- **Method:** Sequence to Sequence functions
- **Implementation:** Go programs & Nextflow pipeline
- **Results:** Identified functions that work accross $\neq$ references
- **Impact:** Used for compression in HiFi assembler¹

keywords: Data Analysis, Pipelining, Machine Learning

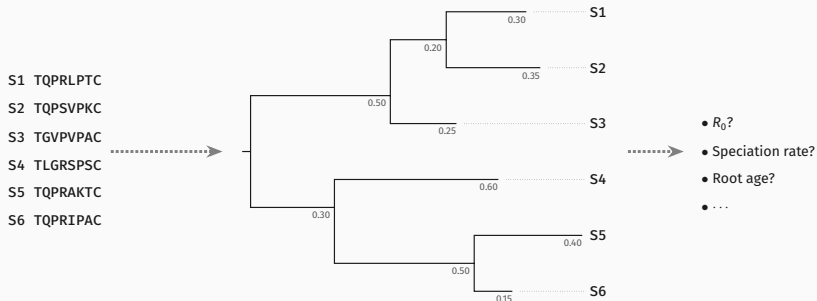


- **Goal:** Find new drug resistance mutations in HIV
- **Method:** Train interpretable classifiers on *treated/non-treated*
- **Implementation:** Python scripts & Snakemake pipeline
- **Results:** Identified new mutations

Blassel et al. 2021 PLoS CB & 2021 Curr. Op. in Vir

Research project - Likelihood-free phylogenetics

Project - Context, phylogenetics



Foundational task in bioinformatics

Citations¹:

NJ: 77k

PhyML: 41k

FastME: 20k

IQTree: 43k

RaxML: 36k

FastTree: 22k

¹From Google scholar

Project - Context, phylogenetic inference

x : **sequence** alignment θ : **tree** M : sequence **evolution model**

Inference:

Max. Likelihood: $\theta^* = \underset{\theta}{\operatorname{argmax}} \mathbf{p(x|\theta, M)}$

Bayesian: $p(\theta|x, M) \propto \mathbf{p(x|\theta, M)}p(\theta)p(M)$

Problem:

Relies on **repeated** estimations of **Likelihood**:

1. **Costly** and slow **computation**: $\mathcal{O}(ns^3 + nLs^2)^1$
2. **Limits** model **realism & complexity**

n : nb. of leaves, L : length of sequence, s : size of alphabet

¹Bryant et al. 2005

Project - Overview

Goal:

Develop **likelihood-free** phylogenetic **methods**

Enable the use of **richer, more realistic** models for phylogenetic tasks

Project Roadmap:

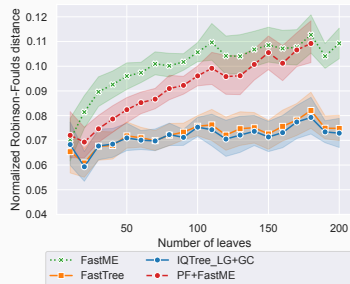
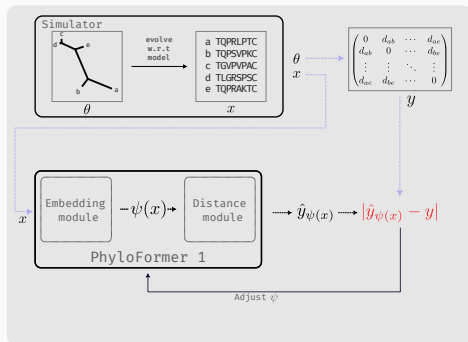
1) Tree inference

- **Alignment-free**
- Tree-space diffusion
- Tree-sequences

2) Inf. conditioned on tree

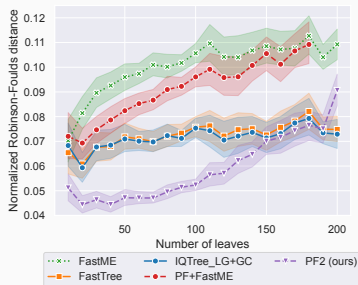
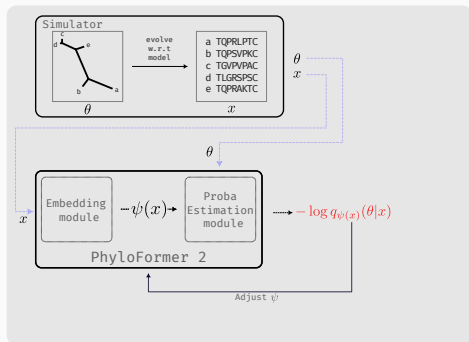
- **Ancestral reconstruction**
- Phylodynamics
- Sequence alignment

Research Project - initial attempts, Post-doc



- Use **distance-matrix** as tree **proxy**
- **Inference:** use **existing** distance **methods** on $\hat{y}_{\psi(x)}$

Nesterenko*, Blassel* et al. 2025, MBE



- **Training:** $\psi^* = \underset{\psi}{\operatorname{argmin}} - \sum_{i \in \text{train}} \log q_{\psi(x_i)}(\theta_i|x_i)$
- **Inference:** we **sample** from $q_{\psi^*(x_E)}(\theta_E|x_E)$

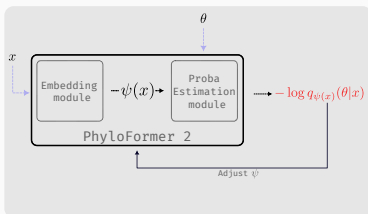
Blassel et al. 2025, preprint

Research Project - Extending NPE to alignment-free

Project - Why alignment-free phylogenetics ?

Problem

- **Multiple** sequence **alignment** (MSA) is **hard**
- **Uncertainty** in MSA can lead to **different trees** and **conclusions**¹



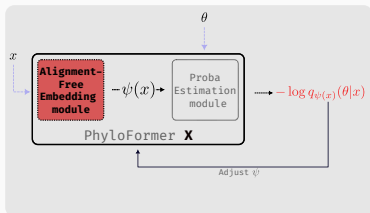
- **PF2** results are **promising**
- **High** potential **impact**

¹Liu et al. 2011; Wong et al. 2008

Project - Why alignment-free phylogenetics ?

Problem

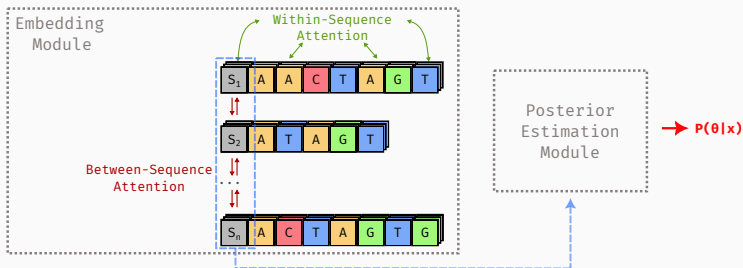
- **Multiple** sequence **alignment** (MSA) is **hard**
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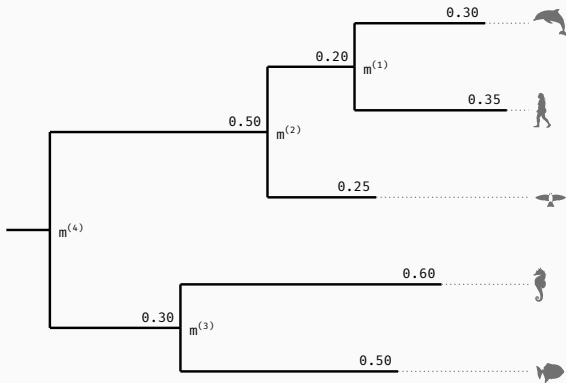
- **PF2** results are **promising**
- **High** potential **impact**
- “Just” need a **new Embedding** method

¹Liu et al. 2011; Wong et al. 2008

Project - An alignment-free embedder

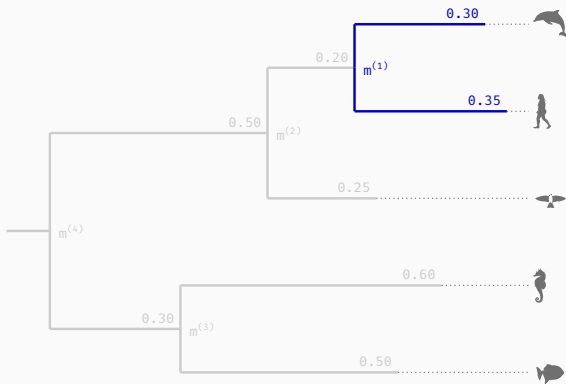


Project - Estimating the Posterior



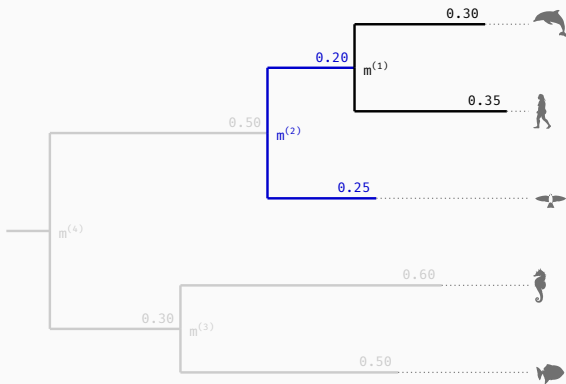
$$q_{\psi(x)}(\theta|x) = ?$$

Project - Estimating the Posterior



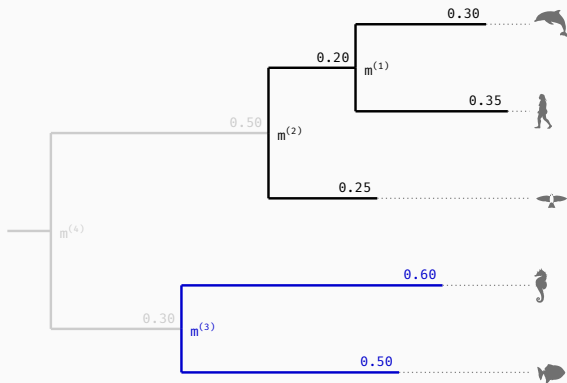
$$q_{\psi(x)}(\theta|x) = p(m^{(1)})$$

Project - Estimating the Posterior



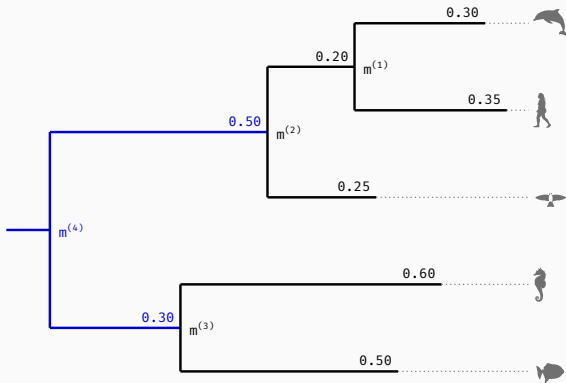
$$q_{\psi(x)}(\theta|x) = p(m^{(1)}) \times p(m^{(2)}|m^{(1)})$$

Project - Estimating the Posterior



$$q_{\psi(x)}(\theta|x) = p(m^{(1)}) \times p(m^{(2)}|m^{(1)}) \times p(m^{(3)}|m^{(1,2)})$$

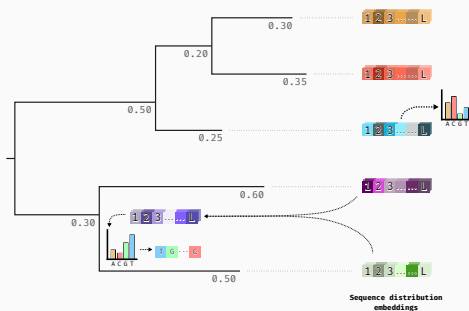
Project - Estimating the Posterior



$$q_{\psi(x)}(\theta|x) = p(m^{(1)}) \times p(m^{(2)}|m^{(1)}) \times p(m^{(3)}|m^{(1,2)}) \times p(m^{(4)}|m^{(1,2,3)})$$

Research Project - Ancestral sequence reconstruction conditioned on a tree

Project - Joint Tree/Parameter learning, ASR



- **per-site distribution** over alphabet
- *Training:* we have **simulated ancestral** seq.
- *Inference:* we **sample** a seq. at **each node**

PF2 presents **posterior calibration** issues

⇒ More **flexible posterior** modelling with **diffusion**

Alignments can still be **useful**

⇒ **Integrate** alignment-free approach **to MSA** pipeline

Can we model **more complex** evolutionary **processes**?

⇒ Infer **tree-sequences** to start tackling **recombination**

Scaling to large number of sequences:

- **17M** SARS-CoV-2 genomes in GISAID
- **900k** genomes in GTDB
- **250k** genomes in IGM database

Extend from **trees** to **networks**:

- **Horizontal transfer** is common in **bacteria**¹
- **Recombination** is frequent in **viruses**²
- *Collaboration with UW-Madison*

numbers: [GISAID](#), [GTDB](#), [IMG](#)

¹Arnold et al. 2022, ²Pérez-Losada et al. 2015

Integration & current collaborations

Analytical Genomics team @ CQSB:

Expertise: SBI/ML (L. Jacob), Evolution (M. Weigt, A. Carbone)

Collaborators: L. Jacob & P. Barrat-Charlaix, M. Weigt (*planned*)

Bioinfo team @ LISN:

Expertise: SBI/ML (F. Pouyet, F. Jay), Trees/Graphs (L. Planche)

Collaborators: F. Pouyet & F. Jay (*planned*)

Modelling of Pathogens team @ IBENS & IP:

Expertise: Evolution & Modelling (A. Zhukova, H. Morlon)

Collaborators: A. Zhukova

Thank you for listening!

I will now gladly answer your questions.

N'hésitez pas à poser vos questions en Français.

Personal Publications

- Lemoine, F. et al. (2020). **COVID-Align: Accurate Online Alignment of hCoV-19 Genomes Using a Profile HMM**. In: *Bioinformatics* 37.12, pp. 1761–1762.
- Blassel, L., A. Tostevin, et al. (2021). **Using Machine Learning and Big Data to Explore the Drug Resistance Landscape in HIV**. In: *PLOS Computational Biology* 17.8, e1008873.
- Blassel, L., A. Zhukova, et al. (2021). **Drug Resistance Mutations in HIV: New Bioinformatics Approaches and Challenges**. In: *Current Opinion in Virology* 51, pp. 56–64.
- Zhukova, A. et al. (2021). **Origin, Evolution and Global Spread of SARS-CoV-2**. In: *Comptes Rendus. Biologies* 344.1, pp. 57–75.
- Blassel, L., P. Medvedev, et al. (2022). **Mapping-Friendly Sequence Reductions: Going beyond Homopolymer Compression**. In: *iScience*, p. 105305.
- Blassel, L., B. Boussau, et al. (2025). **Likelihood-Free Inference of Phylogenetic Tree Posterior Distributions**.
- Nesterenko, L. et al. (2025). **Phyloformer: Fast, Accurate, and Versatile Phylogenetic Reconstruction with Deep Neural Networks**. In: *Molecular Biology and Evolution* 42.4, msaf051.
- Garot, V. et al. (2026). **Teddy: Neural Inference of Epidemiological Parameters from Viral Sequences**.

References

- Arnold, B. J. et al. (2022). **Horizontal Gene Transfer and Adaptive Evolution in Bacteria.** In: *Nature Reviews Microbiology* 20.4, pp. 206–218.
- Bryant, D. et al. (2005). **Likelihood Calculation in Molecular Phylogenetics.** In: *Mathematics of Evolution and Phylogeny*. Ed. by O. Gascuel. Oxford University PressOxford, pp. 33–62.
- Faure, R. et al. (2025). **Alice: Fast and Haplotype-Aware Assembly of High-Fidelity Reads Based on MSR Sketching.** Pre-published.
- Liu, K. et al. (2011). **Multiple Sequence Alignment: A Major Challenge to Large-Scale Phylogenetics.** In: *PLoS Curr* 2, RRN1198.
- Pérez-Losada, M. et al. (2015). **Recombination in Viruses: Mechanisms, Methods of Study, and Evolutionary Consequences.** In: *Infect Genet Evol* 30, pp. 296–307.
- Wong, K. M. et al. (2008). **Alignment Uncertainty and Genomic Analysis.** In: *Science* 319.5862, pp. 473–476.

Supplementary slides

Project - Assembly-free phylogenetic inference

Schema

Project - Tree-sequence inference

frame

Project - Inference on trees, GNNs for R_0

frame

Project - Joint tree-parameter inference, dating

env